

Assignment 2

SDG-Based Questions-CO2

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Topic:

Compare the perceptron convergence theorem with real-world limitations when applied to predicting flood-risk zones from rainfall and elevation data. (SDG 13-Climate Action)

Introduction

The Perceptron Convergence Theorem states that a perceptron learning algorithm will successfully find a separating hyperplane in a finite number of iterations only if the training data is linearly separable. In flood-risk prediction, machine learning models use features such as rainfall, elevation, river level, soil type, drainage, and land use to classify regions as flood-prone or safe. However, real-world flood data is highly complex and often violates the assumptions of the perceptron theorem.

Perceptron Convergence Theorem

- A perceptron is a supervised learning algorithm used for binary classification.
- It updates its weights whenever a training sample is misclassified.
- The theorem guarantees convergence only when the data is linearly separable.
- If a linear boundary exists, the perceptron will find it after a finite number of weight updates.
- If the data is not linearly separable, the algorithm may never converge.

Application to Flood-Risk Prediction

In flood-risk prediction, rainfall and elevation data are collected from weather stations, satellites, and geographic information systems (GIS). The model classifies an area as either:

- Flood-prone
- Non-flood-prone

The aim is to provide early warnings and support disaster management, contributing to SDG 13 – Climate Action.

Comparison of Perceptron Theorem and Real-World Limitations

Perceptron Convergence Theorem	Real-World Flood Prediction Limitations
Assumes data is linearly separable.	Flood-risk data is rarely linearly separable because many environmental factors interact.
Guarantees convergence if a separating boundary exists.	No guarantee of convergence due to noisy and overlapping data.
Uses a single linear decision boundary.	Flood prediction often requires complex non-linear decision boundaries.
Suitable for binary classification.	Flood risk is often classified as low, medium, high, or severe.
Uses a small number of input features.	Real systems require rainfall, elevation, river level, soil moisture, drainage, vegetation, and land use.
Assumes static training data.	Climate change causes rainfall patterns to change over time, requiring continuous retraining.
Performs well on simple datasets.	Real-world datasets contain missing values, sensor errors, and outliers.
Fast and computationally efficient.	More advanced algorithms are needed despite higher computational cost.

Real-World Challenges

1. Non-linear relationships between rainfall and flooding.
2. Climate change causes unpredictable rainfall patterns.
3. Incomplete or noisy data from sensors affects prediction accuracy.
4. Geographical variations make one model unsuitable for all regions.
5. Multiple influencing factors such as river flow, drainage systems, soil type, and urbanization must be considered.

Better Machine Learning Models

To overcome these limitations, the following models are widely used:

- Artificial Neural Networks (ANN)
- Multilayer Perceptron (MLP)
- Random Forest
- Support Vector Machine (SVM)
- Decision Tree
- Deep Learning Models

These models can capture complex, non-linear relationships and provide higher prediction accuracy.

Role in SDG 13 – Climate Action

- Accurate flood-risk prediction helps to:
- Provide early flood warnings.
- Reduce loss of life and property.
- Improve disaster preparedness.
- Support sustainable urban planning.
- Strengthen climate resilience through data-driven decision-making.

Conclusion

The Perceptron Convergence Theorem provides a strong theoretical foundation for binary classification but is effective only when data is linearly separable. In real-world flood-risk prediction, rainfall and elevation data are complex, noisy, and influenced by many environmental factors. Therefore, advanced machine learning techniques such as ANN, Random Forest, SVM, and Deep Learning are preferred for accurate flood prediction, helping achieve SDG 13 – Climate Action through improved disaster management and climate resilience.